Harrison Crosse

Arlington, VA, US harrison@crosse.dev 703-472-7202 hcrosse hcrosse



EMPLOYMENT

Docker

Senior Software Engineer 2025 - Present

- Co-designed Marlin, the event and audit log platform behind Docker's product analytics and AI governance audit logs, on Go, Kafka, and Iceberg.
- Led the backend for Audit Insights, Docker's AI governance analytics, through launch, then designed an exactly-once streaming rewrite using ClickHouse.
- Raised audit log ingestion throughput 8× per worker by batching and pipelining Postgres writes.
- Built Marlin's browser, Node.js, and Rust SDKs on shared protobuf data contracts and moved Docker's web and inference services onto them, which let Docker retire Heap and cut Segment volume >75%.
- Designed and built automatic, schema-driven PII management for Marlin, part of Docker's ISO 27701 certification.
- Cut Iceberg commit retries more than 200× in production by writing data files once and retrying only the metadata commit on conflict.
- Deployed the Flink infrastructure for real-time risk scoring of AI agent activity, and took its alerting service from design to a working demo in less than a week.

Calendly

Senior Data Engineer 2024 - 2025

Data Engineer 2022 - 2024

- Built Calendly's data platform from the ground up: Redshift to BigQuery migration, Airflow, CI/CD, data quality, and developer tooling.
- Ran production Flink pipelines on Kubernetes over Debezium CDC to power AI-transcribed meeting notes, with 6TB of state and peaks of 50k records/s.
- Replaced spreadsheet-defined schemas with GitOps data contracts for hundreds of Segment events, and reprocessed 50TB of replays in Spark.

Amobee

Software Engineer, Data Systems 2021 - 2022

- Moved clickstream processing from batch Spark to Structured Streaming, and Spark from EMR to Kubernetes, eliminating EMR service charges.

Sayari

Data Engineer 2020 - 2021

- Built scraping and fastText NER pipelines for corporate registry data, and migrated infrastructure from GCP to AWS GovCloud with Terraform.

Booz Allen Hamilton

Data Scientist 2019 - 2020

- Modeled Marine Corps unit performance for the Inspector General using clustering and gradient-boosted trees.

EDUCATION

University of Mary Washington

B.S. Computer Science 2019

Minor in Data Science

SELECTED WORK

DataFusion / Ballista

Upstreamed non-blocking distributed shuffle execution by exposing async batch partitioning in DataFusion and offloading Ballista shuffle writes from query workers.

DataFusion #21341 Ballista #1537

Apache Iceberg / Arrow Go

Fixed how iceberg-go detects and retries commit conflicts, added Iceberg's Parquet page-version property, and made Arrow Go's Parquet readable by Snowflake.

iceberg-go #2100 Iceberg #15700 Arrow Go #723

TOOLS

LANGUAGES

Bash Go Java Python Rust SQL TLA+ TypeScript

DATA SYSTEMS

Airflow Arrow BigQuery ClickHouse DataFusion dbt Debezium Flink Iceberg Kafka Parquet PostgreSQL Protobuf Snowflake Spark

INFRASTRUCTURE

Argo AWS Docker GCP GitHub Actions Grafana Helm Kubernetes Linux Terraform